Interpreting Deep Networks



1st Case: Adversarial Examples

Deep networks (and most other machine learning methods) **are easily fooled!**

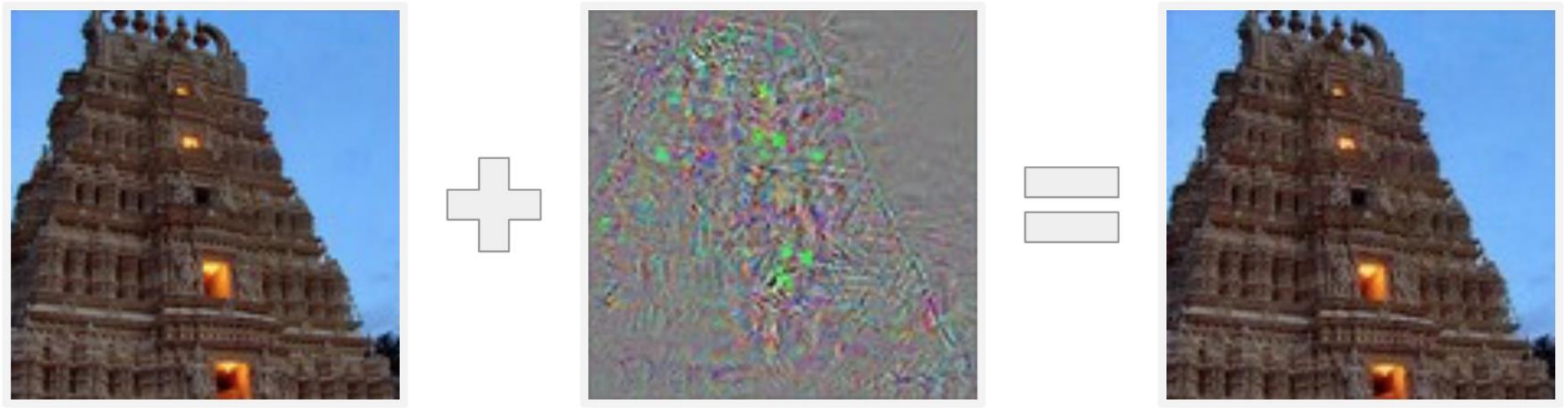
$$\max_{\{\boldsymbol{\epsilon} \in \mathbb{R}^d \,|\, \|\boldsymbol{\epsilon}\|_p < \mathrm{C}\}} distance(f_{\boldsymbol{\theta}}(\mathbf{x}), f_{\boldsymbol{\theta}}(\mathbf{x} + \boldsymbol{\epsilon}))$$

For instance: $\boldsymbol{\epsilon} = \lambda \frac{d\, l(f_\theta(\mathbf{x}), \mathrm{y}_i)}{d\, \mathbf{x}}\Big|_{\mathbf{x}_i}$

[Szegedy et al., "Intriguing properties of neural networks", NIPS 2014]

1st Case: Adversarial Examples

Deep networks (and most other machine learning methods) **are easily fooled!**



Original image

Temple (97%)

Perturbations

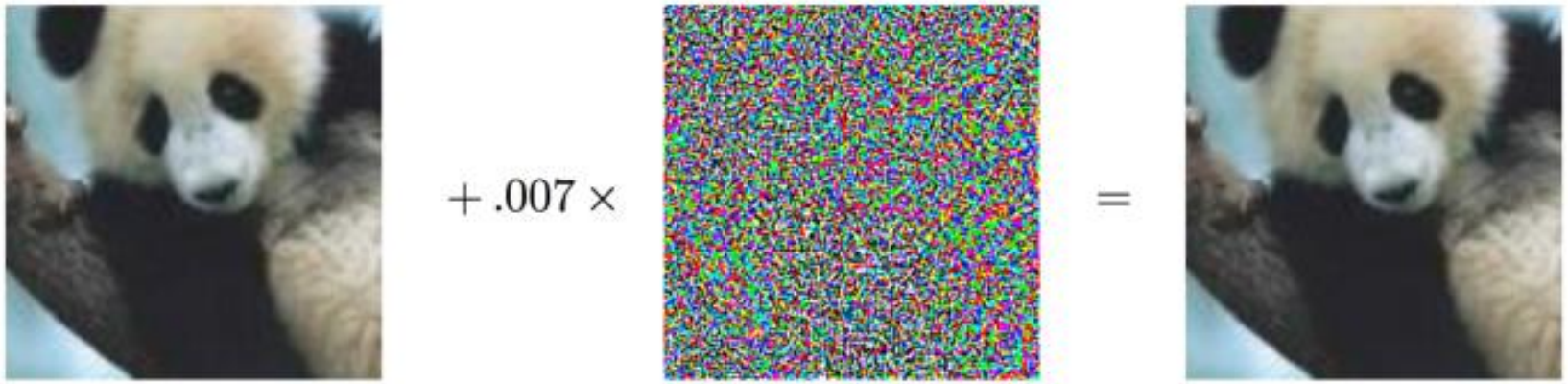
Adversarial example

Ostrich (98%)

[Szegedy et al., "Intriguing properties of neural networks", NIPS 2014]

1st Case: Adversarial Examples

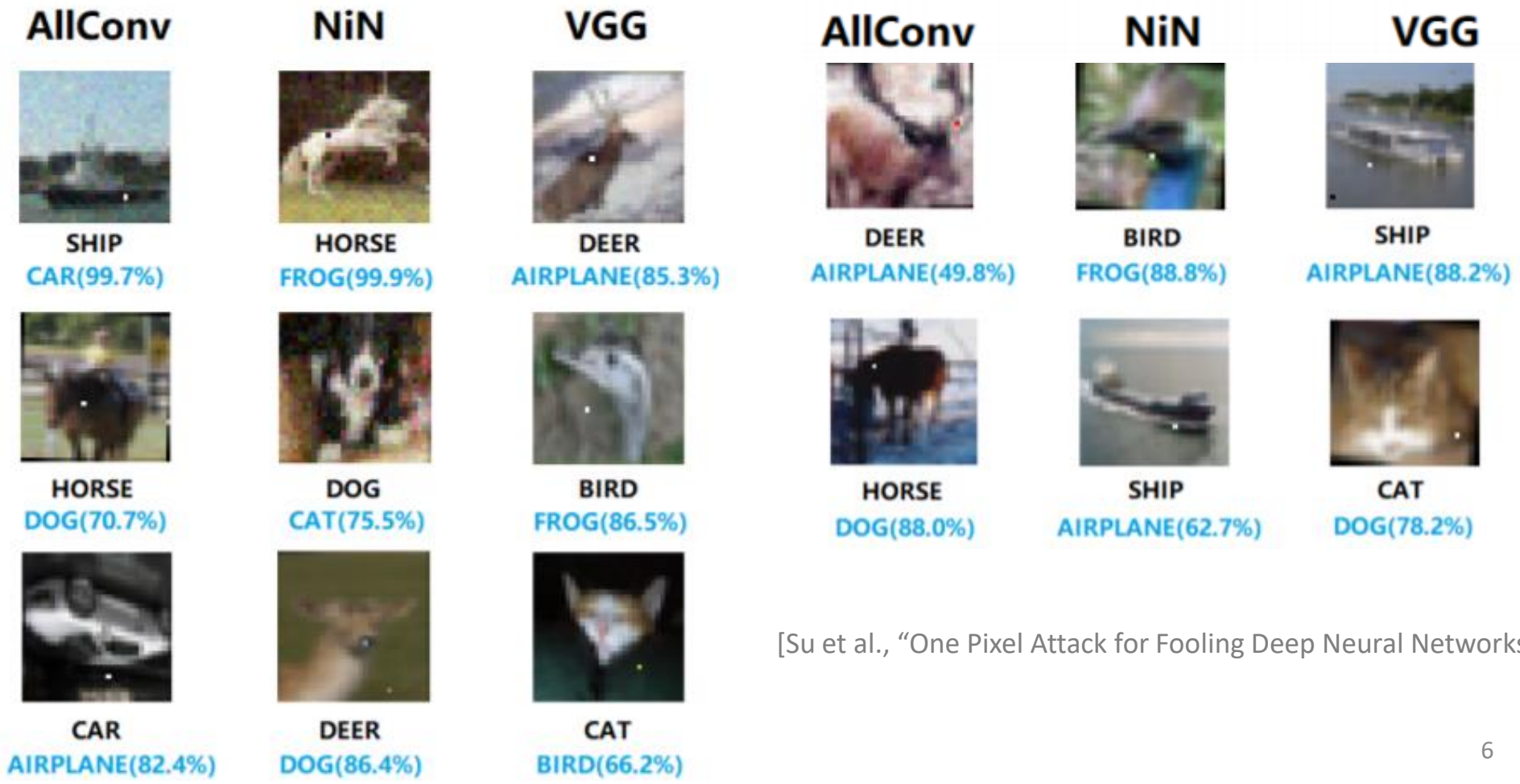
Deep networks (and most other machine learning methods) **are easily fooled!**



[Szegedy et al., "Intriguing properties of neural networks", NIPS 2014]

1st Case: Adversarial Examples

Deep networks (and most other machine learning methods) **are easily fooled!**



[Su et al., "One Pixel Attack for Fooling Deep Neural Networks", 2019]

1st Case: Adversarial Examples

Deep networks (and most other machine learning methods) **are easily fooled!**



Cup(16.48%)
Soup Bowl(16.74%)



Bassinet(16.59%)
Paper Towel(16.21%)



Teapot(24.99%)
Joystick(37.39%)



Hamster(35.79%)
Nipple(42.36%)

[Su et al., "One Pixel Attack for Fooling Deep Neural Networks", 2019]

1st Case: Adversarial Examples

Deep networks (and most other machine learning methods) **are easily fooled!**

- 66.97% of the images in Kaggle CIFAR10 dataset can be fooled with 1-pixel modifications.
- 16.04% images in the ImageNet dataset can be fooled with 1-pixel modifications.

[Su et al., “One Pixel Attack for Fooling Deep Neural Networks”, 2019]

1st Case: Adversarial Examples

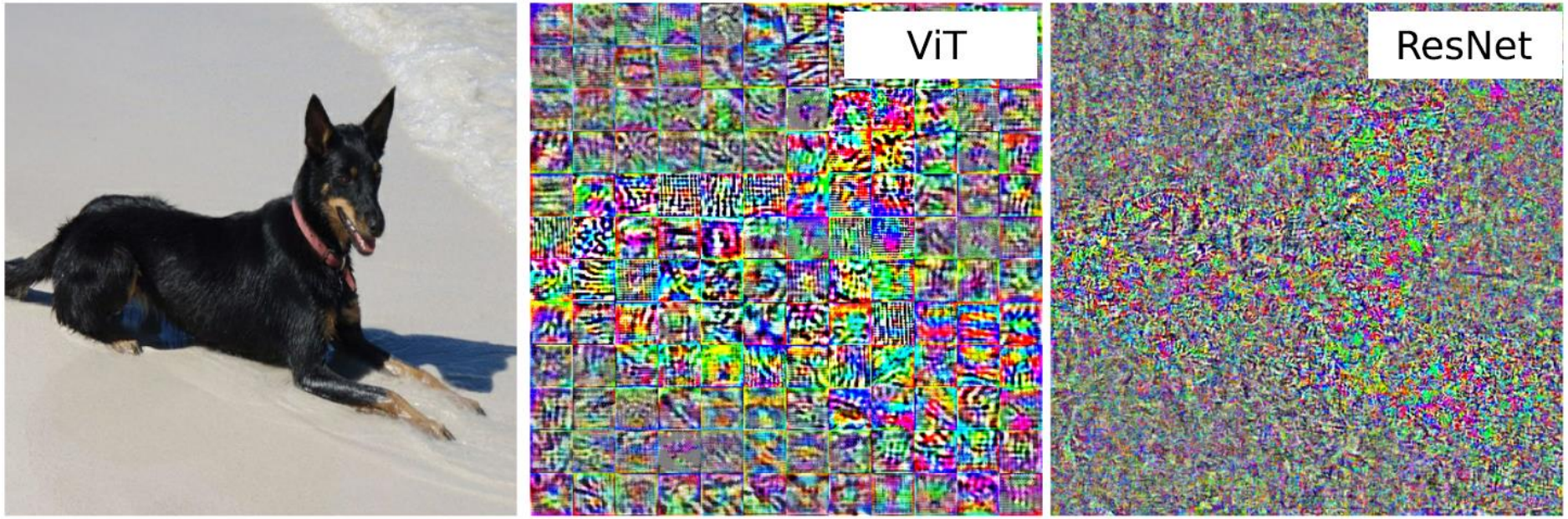
Adversarial Attacks can transfer from one (ConvNet) model to another

	VGG16	InceptionV3	InceptionV4	Inception ResNet-V2	ResNet-V2	Ensemble
No perturbation	86.8%	96.4%	97.6%	100%	89.6%	99.8%
Random noise	81.3%	91.7%	94.6%	97.8%	84.5%	98.1%
VGG16-TAP	**3.2%**	**23.9%**	28.1%	32.3%	**23.9%**	26.7%
VGG16-FGSM	3.7%	34.9%	44.0%	50.0%	34.7%	46.4%
VGG16-BIM	4.0%	24.2%	**24.5%**	**28.5%**	**23.9%**	**22.7%**
VGG16-UAP	12.4%	31.2 %	32.8%	46.9 %	33.2%	43.7%
Inc-V3-TAP	**29.4%**	**0.0%**	**22.1%**	**24.7%**	**46.9%**	**30.8%**
Inc-V3-FGSM	57.7%	26.9%	70.2%	72.9%	65.7%	75.4%
Inc-V3-BIM	66.0%	0.01%	67.7%	70.2%	76.8%	73.6%
Inc-V3-UAP	39.8%	52.2%	56.4%	63.1%	50.5%	64.6%
Inc-V3-MI-FGSM	45.9%	0.1%	47.3%	50.7%	61.8%	62.5%
Inc-V3-CW	84.9%	24.5%	93.5%	98.6%	86.9%	96.9%

[Zhou et al, "Transferable Adversarial Perturbations", ECCV 2018]

1st Case: Adversarial Examples

Adversarial Attacks might not transfer from one (ConvNet) model to another (visual transformer)

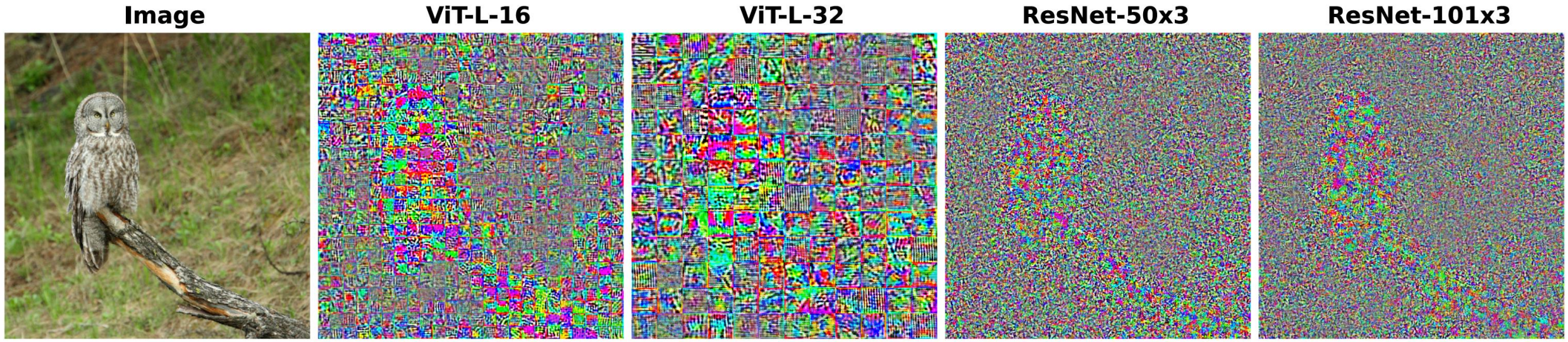


Adversarial perturbations computed for a Transformer and a ResNet model, which are qualitatively quite different.

[Bhojanapalli et al, "Understanding Robustness of Transformers for Image Classification", ICCV 2021]

1st Case: Adversarial Examples

Adversarial Attacks might not transfer from one (ConvNet) model to another (visual transformer)



Adversarial perturbations computed for a Transformer and a ResNet model, which are qualitatively quite different.
Depth/Width show negligible qualitative difference.

[Bhojanapalli et al, "Understanding Robustness of Transformers for Image Classification", ICCV 2021]

1st Case: Adversarial Examples

Adversarial Attacks might not transfer from one (ConvNet) model to another (visual transformer)

	ViT→RN	RN→ViT
ViT-B/16 vs. RN-101x1	79.7% (-2.5)	85.2% (-0.4)
ViT-B/32 vs. RN-50x3	82.2% (-1.7)	80.9% (-0.3)
ViT-L/16 vs. RN-101x3	84.3% (-1.8)	85.8% (-0.6)
ViT-L/32 vs. RN-152x4	85.4% (-0.7)	86.5% (-0.2)

on average adversarial examples do not significantly hurt performance when transferred between transformers and ConvNets.

[Bhojanapalli et al, "Understanding Robustness of Transformers for Image Classification", ICCV 2021]

We might need to understand how deep network's work in general or per example to make them robust to such adversarial examples

Methods to alleviate these are studied in the subfield of "defense against adversarial attacks"



2nd Case: Bias and Fairness of Models

Standard ML applications can be “unfair”, “unjust”, or “discriminatory”

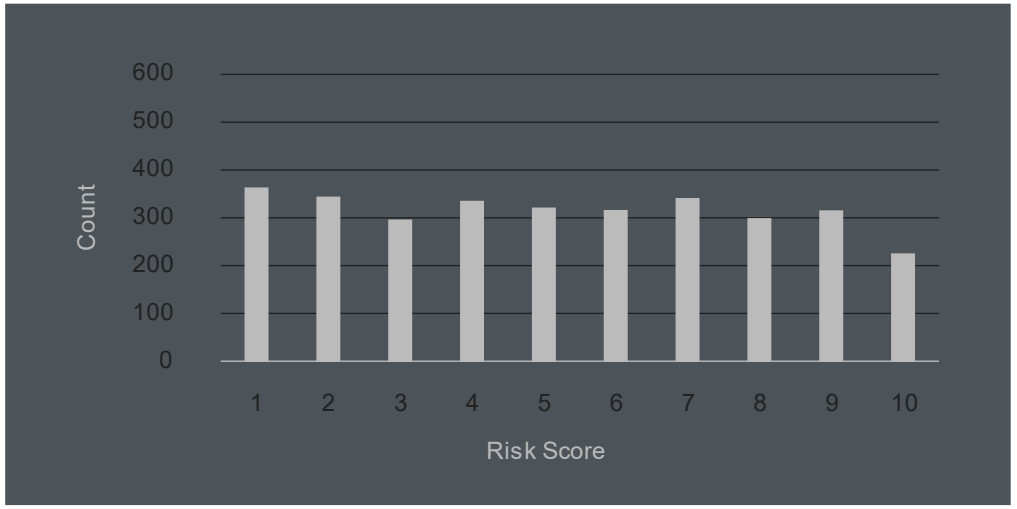


[source: https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing]

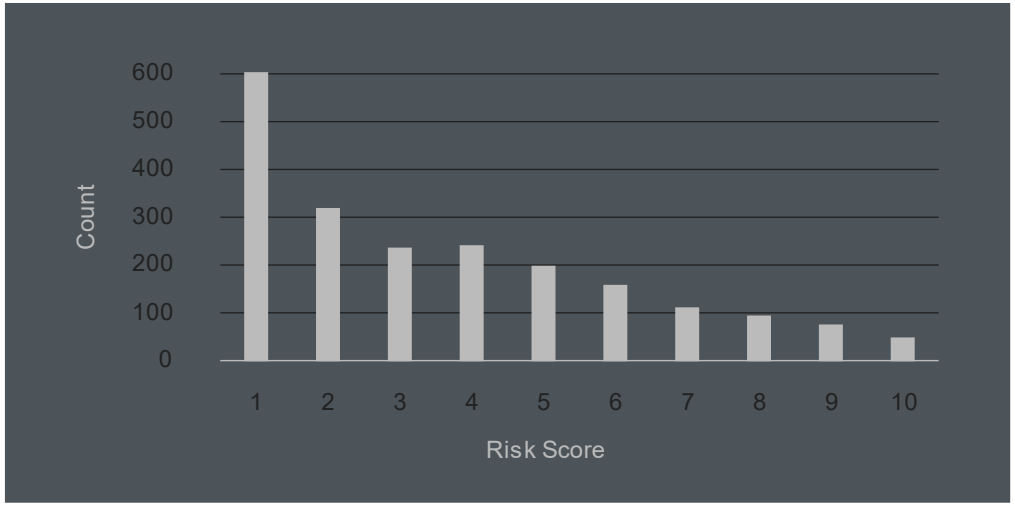
2nd Case: Bias and Fairness of Models

Standard ML applications can be “unfair”, “unjust”, or “discriminatory”

Risk Score Distribution for Black Defendants



Risk Score Distribution for White Defendants



[source: https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing]

We sometimes need to analyze how deep network's work in general and monitor how they make individual decisions to ensure fairness

Methods to alleviate these issues are studied in the subfield of "ML Fairness". Also slightly discussed in the lecture on "ML best research practice"

With these in mind,

- what do deep networks use to make their prediction/estimation/decision?

- can we trust in neural network's prediction in mission-critical applications?

The family of methods that try to interpret trained models will help answer such questions

